

User Incident



Planner Agent



Log / Config Tools



Analyzer Agent



Validator (Critic)



Human Approval



Final Recommendation

ai-incident-advisor/

|

├─ agent.py # Main workflow

├─ tools/

| └─ log_reader.py

| └─ config_reader.py

|

├─ data/

| └─ logs/app.log

| └─ configs/deployment.yaml

|

├─ schemas/

| └─ incident_report.json

|

├─ .env

├─ requirements.txt

└─ README.md

- **Engineering discipline**
- **LLM safety patterns**
- **Agentic reasoning**
- **Production thinking**
- **Not just prompt hacking**

This project shows LLM workflow patterns—

- ✚ context injection,
- ✚ explicit planning,
- ✚ tool-augmented reasoning,
- ✚ structured outputs,
- ✚ validation loops, and
- ✚ human-in-the-loop controls

used together to build a reliable agentic system rather than a simple chatbot

```
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-aiagents-portfolio-amith\agentic-ai-openai\p4aiincidentadvisor> ls

Directory: C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-aiagents-portfolio-amith\agentic-ai-openai\p4aiincidentadvisor

Mode                LastWriteTime         Length Name
----                -
d-----          30-12-2025   16:06           data
d-----          30-12-2025   16:00          schemas
d-----          30-12-2025   16:00           tools
-a-----          30-12-2025   16:04           179 .env
-a-----          30-12-2025   16:03           31 requirements.txt
```

```
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agentical-ai-agents-portfolio-amith\agentical-ai-openai\p4aiincidentadvisor> python agent.py
PLAN:
'''json
{
  "stepByStepPlan": [
    {
      "step": 1,
      "action": "Identify the error in the application logs",
      "expectedOutcome": "Recognize that the application failed to connect to the database on db:5432."
    },
    {
      "step": 2,
      "action": "Confirm database service status",
      "expectedOutcome": "Make sure that the database service is up and running on the expected host and port."
    },
    {
      "step": 3,
      "action": "Check the value of DB_HOST in the deployment configuration",
      "expectedOutcome": "DB_HOST should be set to 'db', which means an internal service name."
    }
  ]
}
```

```

    }
  }
}
...
}

[+] INCIDENT REPORT:

{
  "incident": "Application failed to connect to the database leading to startup failure.",
  "root_cause": "The application is unable to connect to the database at db:5432 due to a connection refused error.",
  "severity": "high",
  "recommended_fix": "Investigate the database service to ensure it is running and configured correctly to accept connections on the expected host and port."
,
  "confidence": 0.9
}

Approve recommendation? (yes/no): yes
[+] Recommendation approved (no action executed - advisory only)

```

```

Recommendation approved (no action executed - advisory only)
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-aiagents-portfolio-amith\agentic-ai-openai\p4aiincidentadvisor> python agent.py

PLAN:
{
  "stepByStepPlan": [
    {
      "step": 1,
      "action": "Review application logs for connection errors",
      "details": "Analyze the 'ERROR Connection refused at db:5432' message to understand the connection failure to the database."
    },
    {
      "step": 2,
      "action": "Verify database service availability",
      "details": "Check if the database service is up and running. This involves confirming that the database is deployed and accessible."
    },
    {
      "step": 3,
      "action": "Check network accessibility",
      "details": "Ensure that the application can reach the database using the specified DB_HOST and DB_PORT. This includes checking network policies and service configurations."
    },
    {
      "step": 4,
      "action": "Inspect database configuration",

```

```
}
[✓] INCIDENT REPORT:

{
  "incident": "Connection refused error at db:5432 leading to application failure",
  "root_cause": "The database service is unavailable or incorrectly configured, which prevents the application from establishing a connection.",
  "severity": "high",
  "recommended_fix": "Verify that the database service is running and configured correctly, including checking network accessibility and deployment configurations.",
  "confidence": 0.85
}

Approve recommendation? (yes/no): no
[✗] Action rejected by human
```